Google models such as gemini-1.5-pro-002 (Reid et al., 2024), and Mistral models such mistral-large-2407 (Jiang et al., 2023).

For open-source models, we include models such as Llama-3.1-405b-instruct, Llama-3.1-70b-instruct (Dubey et al., 2024), deepseek-v2.5, (Liu et al., 2024a), qwen2.5-72b-instruct (Team, 2024b; Yang et al., 2024), command-r-plus-08-2024 (Cohere, 2024; Cohere For AI, 2024), gemma-2-27b-it (Team, 2024a; Team et al., 2024), mixtral-8x22b-instruct-v0.1 (Jiang et al., 2023), and phi-3.5-moe-instruct (Abdin et al., 2024). See Table 4 for a full list of citations.

For all models and tasks, we perform single-turn evaluation with temperature 0, unless otherwise noted in the model card. All models run with their respective templates from our updated version of FastChat (Zheng et al., 2024). We run all open-source models with bfloat16. When running new models, we take care to set up its hyperparameters and chat template as in the model's example code, and we also double check the outputs to make sure that the inference, as well as our automated parsing functions, are working correctly and fairly. See more details in Appendix A.4 and Appendix A.5. For each question, a model receives a score from 0 to 1. For each model, we compute the score on each task as the average of all questions, we compute the score on each of the six categories as the average of all their tasks, and we compute the final LiveBench score as the average of all six categories. In Appendix B, we give additional documentation including average input/output tokens and cost to run LiveBench for each API model.

3.1 Discussion of Results

We compare all 40 models on LiveBench according to the experimental setup described above; see Table 1 and Table 2. We find that o1-preview-2024-09-12 performs the best overall, 6% better than all other models. o1-preview-2024-09-12 substantially outperforms all other models in the data analysis, language, and math categories. The next-best model is claude-3-5-sonnet-20240620, which far outperforms all other models in the coding category (although o1-mini outperforms claude-3.5 in code generation, claude-3.5 has the edge in code completion). o1-mini-2024-09-12 is third overall and is significantly better than all other models in the reasoning category.

The best-performing open-source models are llama-3.1-405b-instruct and qwen2.5-72b-instruct, which virtually tie with each other and outperform gpt-4-turbo. The best-performing small open-source model is phi-3.5-moe-instruct (see Table 2): with only 6.6B active parameters, it outperforms gpt-3.5 and is on par with mixtral-8x22b.

3.2 Correlation Analyses

Now we present analyses involving correlation among different categories and tasks. First, we compute the Pearson correlation coefficient among all pairs of categories and tasks in LiveBench (see Figure 2). We find that unsurprisingly, math, coding, and reasoning all correlate with one another. Interestingly, language correlates fairly well with data analysis, likely due to both categories including tasks that require the LLM to output a large part of the prompt that is modified in a specific way (e.g., by fixing typos or changing the table format). Surprisingly, instruction following correlates relatively weakly with all other categories. Among tasks, we see that math comp correlates the highest with the average LiveBench performance, suggesting that this task is the greatest indicator of overall model performance. This is likely due to these being high-quality, diverse mathematical reasoning questions (which we modified to reduce contamination).

Next, in order to see the strengths and weaknesses of each model, we create a scatterplot of each model's overall LiveBench performance vs. performance on a single category or task (Figure 3). By plotting a best fit line and computing the residuals for each model, we can compute which models are outliers in specific categories – that is, models that are disproportionately stronger in a particular category relative to the best fit line. We see that the o1 and phi series of models are outliers in terms of reasoning (Figure 3, left), while some of the Llama, gemini, and command-r models are outliers in terms of instruction following. We present additional details in Appendix A.1, including a table of each model's relative best and worst tasks (computed as the highest and lowest residuals).

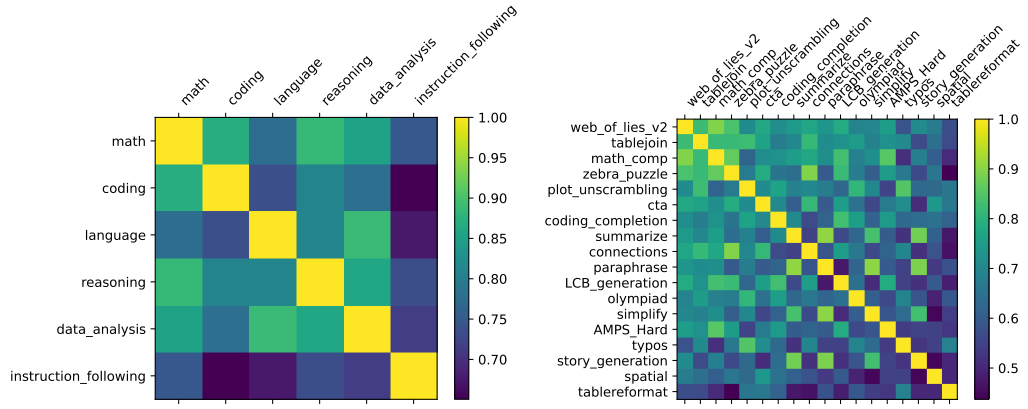

Figure 2: **Correlations among categories and groups in `LiveBench`.** For each pair of categories (left) and tasks (right) in `LiveBench`, we compute the Pearson correlation coefficient based on the results for all 40 models.

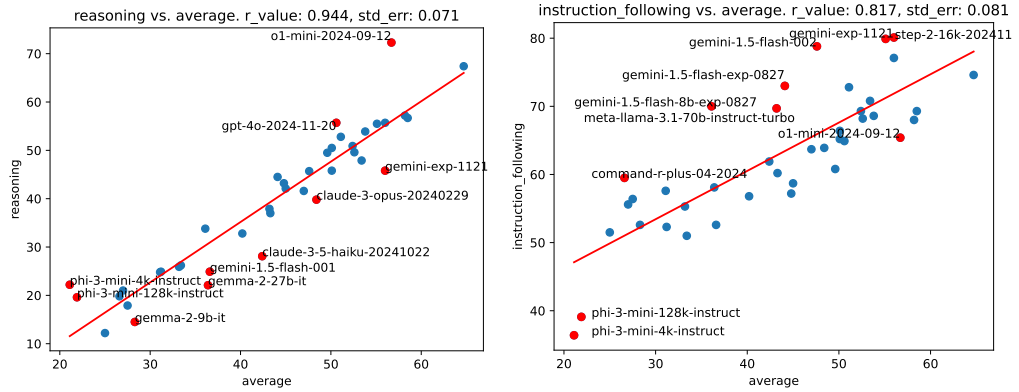

Figure 3: **Reasoning and instruction following performance vs. average performance.** We plot the performance of all 40 models' reasoning (left) and instruction following (right) performance compared to overall `LiveBench` performance, computing a best fit line and annotating outliers.

3.3 Comparison to Other LLM Benchmarks

Next, we compare `LiveBench` to two prominent benchmarks, ChatBot Arena (Chiang et al., 2024) and Arena-Hard (Li et al., 2024). In Figure 4, we show a bar plot comparison among models that are common to both benchmarks, and in Figure 6, we compare the performance of these models to a best-fit line. We also compute the correlation coefficient of model scores among the benchmarks: `LiveBench` has a 0.91 and 0.88 correlation with ChatBot Arena and Arena-Hard, respectively.

Based on the plots and the correlation coefficients, we see that there are generally similar trends to `LiveBench`, yet some models are noticeably stronger on one benchmark vs. the other. For example, `gpt-4-0125-preview` and `gpt-4-turbo-2024-04-09` perform substantially better on Arena-Hard compared to `LiveBench`, likely due to the known bias from using `gpt-4` itself as the LLM judge (Li et al., 2024). We hypothesize that the strong performance of some models such as the `gemini-1.5` models on ChatBot Arena compared to `LiveBench` may be due to having an output style that is preferred by humans. These observations emphasize the benefit of using ground-truth judging, which is immune to biases based on the style of the output.

Comparison between Ground-Truth and LLM-Judging As an additional comparison between `LiveBench` and LLM judge based benchmarks, we give a preliminary study in the Appendix on the efficacy of LLM judging for hard math and reasoning questions. Specifically, we run an initial

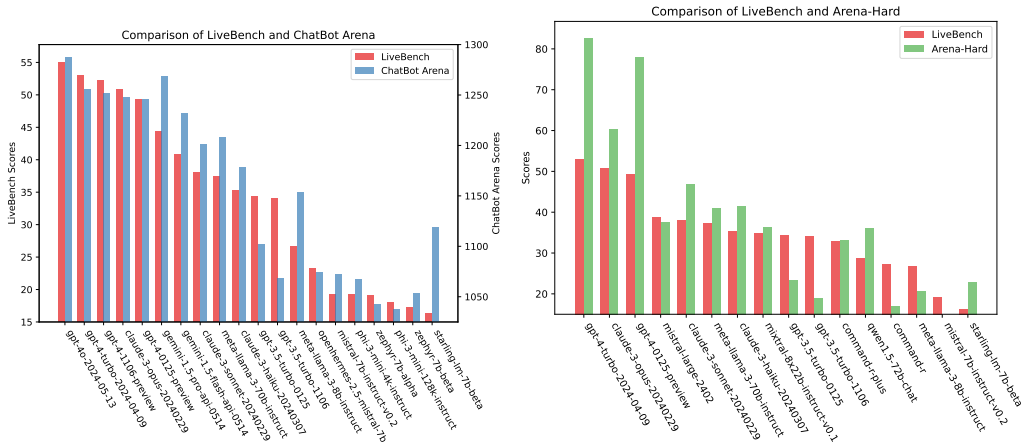

Figure 4: **Comparison of LiveBench to other LLM benchmarks.** We compare `LiveBench` to ChatBot Arena (left) and Arena-Hard (right). We see that while there are generally similar trends, some models are noticeably stronger on one benchmark vs. the other. For example, both GPT-4 models are substantially better on Arena-Hard.

experiment regarding the question, 'if an LLM struggles to answer a hard math or reasoning question, then will the LLM also struggle to determine whether or not a given answer to that question is correct?' Our experiments give evidence that the answer is yes, for zebra puzzles and AMC/AIME questions, but the results are not definitive. See Appendix A.2.

3.4 Analysis of Monthly Updates

As described in Section 2.7, we have completed two monthly updates for `LiveBench` so far. The rank correlation between the original and first update, and the first and second update, are both > 0.997, showing that the model rankings have stayed consistent. On the other hand, between the original and the most-recent set of questions, the median and mean average scores (among models included in all iterations of the leaderboard) have both dropped by about 1.2%, showing that the benchmark is becoming harder over time, as newly released models become more capable.

4 Related Work

We describe the most prominent LLM benchmarks and the ones that are most related to our work. For a comprehensive survey, see (Chang et al., 2024). The Huggingface Open LLM Leaderboard (Gao et al., 2021; Beeching et al., 2023) is a widely-used benchmark suite that consists of static datasets such as Big Bench Hard (Suzgun et al., 2023) and MMLU-Pro (Wang et al., 2024). While this has been incredibly useful in tracking the performance of LLMs, its static nature leaves it prone to test set contamination by models.

LLMs-as-a-judge. AlpacaEval (Li et al., 2023b; Dubois et al., 2023; 2024), MT-Bench (Chiang et al., 2024), and Arena-Hard (Li et al., 2024) are benchmarks that employ LLM judges on a fixed set of questions. Using an LLM-as-a-judge is fast, relatively cheap, and has the flexibility of being able to evaluate open-ended questions, instruction-following questions, and chatbots. However, LLM judging also has downsides. First, LLMs have biases towards their own answers (Li et al., 2024). In addition, GPT-4 judges have a noticeable difference in terms of variance and favorability of other models compared to Claude judges. Additionally, LLMs make errors. As one example, question 2 in Arena-Hard asks a model to write a C++ program, yet GPT-4 incorrectly judges `gpt-4-0314`'s solution as incorrect (Li et al., 2024).

Humans-as-a-judge. ChatBot Arena (Chiang et al., 2024; Zheng et al., 2024) leverages human prompting and feedback. Users ask questions and receive outputs of two randomly selected models and pick which output they prefer. This preference feedback is aggregated into an Elo score for each model. While human evaluation is great for capturing the preferences of a crowd, using a human-